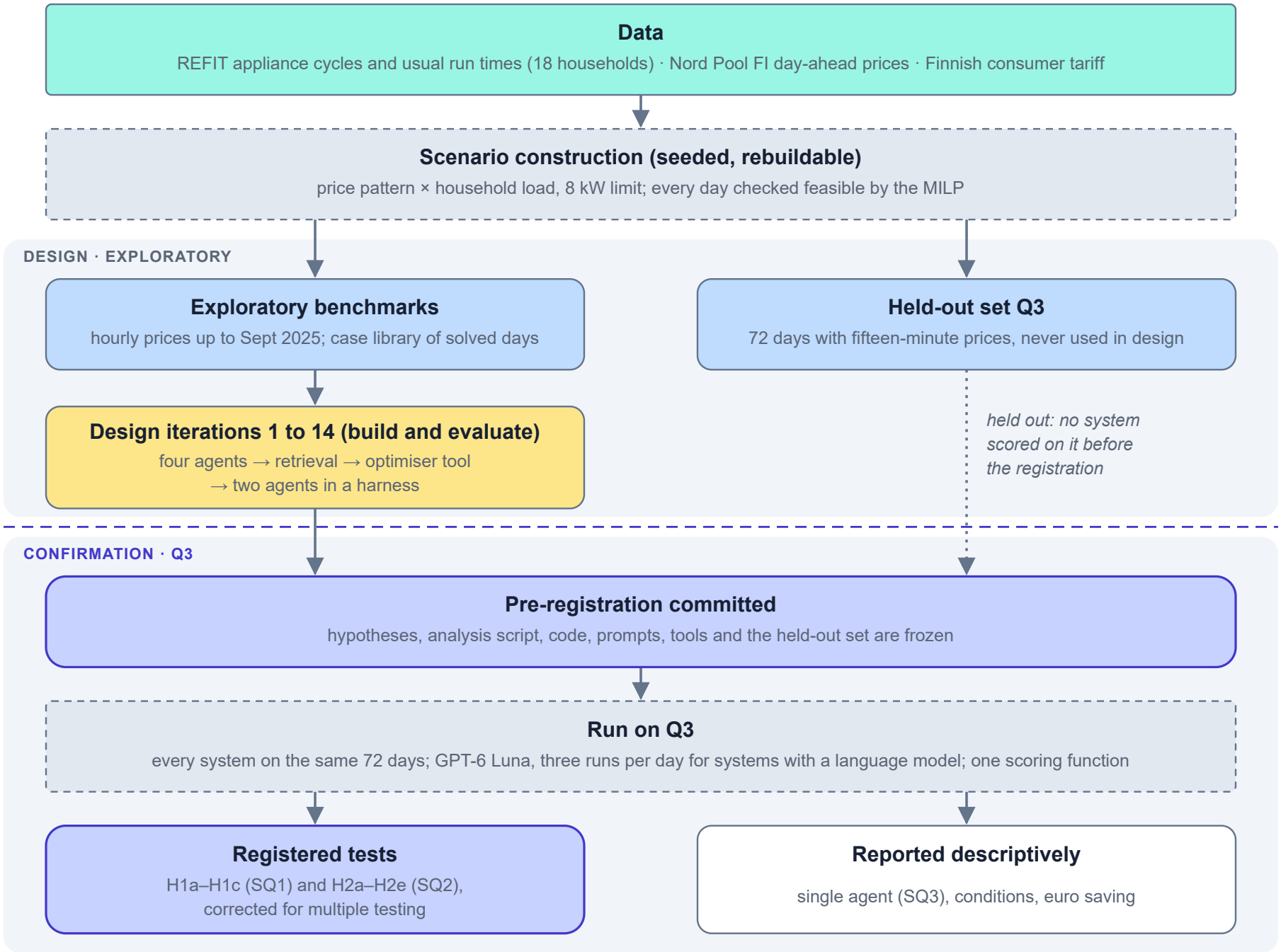

Data
REFIT appliance cycles and usual run times (18 households) · Nord Pool FI day-ahead prices · Finnish consumer tariff
Scenario construction (seeded, rebuildable)
price pattern × household load, 8 kW limit; every day checked feasible by the MILP
DESIGN · EXPLORATORY
Exploratory benchmarks
hourly prices up to Sept 2025; case library of solved days
Held-out set Q3
72 days with fifteen-minute prices, never used in design
Design iterations 1 to 14 (build and evaluate)
four agents → retrieval → optimiser tool
→ two agents in a harness
held out: no system
scored on it before
the registration
CONFIRMATION · Q3
Pre-registration committed
hypotheses, analysis script, code, prompts, tools and the held-out set are frozen
Run on Q3
every system on the same 72 days; GPT-6 Luna, three runs per day for systems with a language model; one scoring function
Registered tests
H1a–H1c (SQ1) and H2a–H2e (SQ2),
corrected for multiple testing
Reported descriptively
single agent (SQ3), conditions, euro saving